

On the Learnability of Linguistic Variation

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1 Background

We start with two observations which we regard as self-evident although some additional remarks will do no harm. The first is that variation permeates all corners of language. While we will not address how such variability is encoded in the speaker’s language — the familiar grammar vs. usage debate (Cedergren and Sankoff 1974, Kay and McDaniel 1979, Sankoff and Labov 1979, and more recently Newmeyer 2003, Bybee 2010) — we stress that the fact of variation in no way denies the reality of categorical constraints on linguistic structures (Labov 1969, Sankoff 1988, Kroch 1989, Yang 2002, Sankoff and Blondeau 2007). Indeed, we will present new arguments that probabilistic linguistic variation necessarily contains an element of categoricity in their representation and acquisition.

Second, we regard a psychological theory of learning as a central component in the study of variation and change, following both traditional and modern literature (Paul 1888, Halle 1962, Sankoff and Laberge 1974, Lightfoot 1979, Labov 1989): variation would be lost if it is not passed on to later generations of speakers. Linguistic knowledge can be propagated by both child and adult language learners (*transmission* and *diffusion*; Labov 2007 and Section 4). For the purpose of this article, we will not address the actuation problem (Weinreich et al. 1968), namely, how linguistic variants arise in a speech community. Our main concern here is how such variants are learned and passed on regardless the manner in which they are introduced. We do not rule out the possibility that language learners may be responsible for actuation. As we review later, there are clearly individual differences in language learning by both children and adults, which may sow the seeds of language variation and change.

To develop a theory of how language variation is propagated over time faces considerable challenges. Some aspects of language are transmitted essentially veridically, resulting in stability in both categorical and probabilistic forms (Labov 1989). In other cases, however, linguistic variants must be transmitted differentially, with one form gradually displacing another, the most visible type of language change, including changes during individual speaker’s lifespan (Sankoff and Blondeau 2007, Sankoff 2019). Furthermore, any theory of variation and its transmission must confront what can be referred to as the enigma of uniformity: “the uniformity of abstract patterns of variation which are invariant in respect to particular levels of usage” (Labov 1972, 120), as well as the uniformity in the direction and outcome of linguistic change (Labov 2007, Labov et al. 2006). Thus, a model of linguistic variation and change must contain elements of stability as well as plasticity: the interaction between these opposing forces must allow for the different outcomes to emerge under suitable internal and external conditions, including the largely invariant grammar characteristic of a speech community.

So far so familiar. In this paper, we highlight yet another important component, and challenge, for a theory of linguistic variation and change. In our own quantitative studies of language (Labov 1981, Roberts

and Labov 1995, Yang 2013, 2016, Labov 2017), we have had to wrestle with the *sparsity* of linguistic data: the critical data that best informs theoretical positions is often the rarest kind. While this observation would be familiar to everyone engaged with corpus work, the sparsity problem takes on new urgency in a psychological setting: How does a language-learning child acquire the general rules of language, including those that show inherent variability, when they are only sparsely represented in the input? We illustrate these problems in Section 2 with several familiar case studies in language variation and acquisition. In Section 3 we propose a solution to the sparsity problem, synthesizing ideas from both formal and empirical studies of language learning and generalization (Labov 1989, Yang 2016, Sankoff 2019). In Section 4 we revisit the model of language transmission and change (Labov 2007) by providing an explicit account of the differences between child and adult language acquisition.

2 Data Sparsity and Linguistic Systematicity

To speak of linguistic transmission, we must be clear about what kind of linguistic information is transmitted. The question, in our view, turns out to a much larger debate that has energized linguistics for generations. In what follows, we describe how our own quantitative work bears on the issue, and why we believe language acquisition must play a much larger role than previously realized.

Our starting point is one of the oldest debates in the study of language change (Labov 1981): In the evolution of sound systems, is the basic unit of change the word or the sound? Osthoff and Brugman 1878 (translated in Lehmann 1967, p204) put the Neogrammarian position in unambiguous terms: “every sound change, inasmuch as it occurs mechanically, takes place according to laws that admit no exception.” In modern linguistics, the opposing view is best represented by the theory of lexical diffusion (Wang 1969, 1979, Phillips 1984, Bybee 2002): “A phonological rule gradually extends its scope of operation to a larger and larger portion of the lexicon, until all relevant items have been transformed by the process” (Chen and Wang 1975, p256).

An empirical assessment of these opposing positions was undertaken with data from ongoing language variation and change (Labov 1981), with a reappraisal in Labov (1994). Here we summarize the results from a most recent study (Labov 2017), which draws from data in the Philadelphia Neighborhood Corpus (Labov et al. 2013, henceforth PNC). One of the characteristic features of the Philadelphia phonological system is the raising of /ey/ in the front diagonal, which selects words with consonants following stressed /ey/ (*made, eight, main, days*) as opposed /ey/ in final position, before hiatus, and liquids (*day, mayor, hale*). The raising of the allophone, designated as /eyC/ here, was first observed in the 1970s. This process is especially suitable for understanding the mechanics of variation and change because it is well below the level of conscious awareness and had not been observed in the previous qualitative descriptions of the Philadelphia dialect (e.g., Tucker 1944).

The PNC corpus contains 28,637 /eyC/ words. Regression analysis of the change shows no effect of social factors but clear effect of phonetic factors. In particular, manner of the coda consonant has considerable influence on the change in progress, in the order stop > affricate > fricative > nasal; see Labov 2017 for details. Here we focus on the contrasting predictions of structural and lexical accounts of language change: Do all words within a phonological class change simultaneously or do they proceed on a one-by-one lexical basis? Figure 1 provides the linear trend lines for the ten most frequent (eyC) words over a 40 year span, ranging from 471 to 1,272 tokens per word. It is evident that these words all move together, showing no evidence of random selection. Furthermore, these trajectories follow the logic of their phonetic composition: initial zero, velars and palatal onsets in *hate, eight, gave, came* lead in the raising process, while velar codas in *take* and *make* are much lower.

An analytical conundrum arises at this juncture. While the most frequent words show regularity of sound change, how can it be demonstrated for words of low frequencies and in the limiting case, those that appear

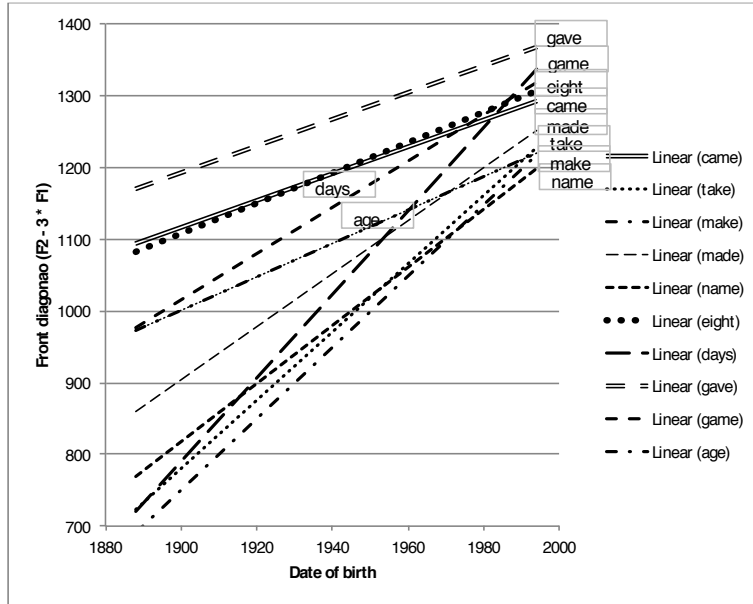


Figure 1: Linear regression trend lines of the ten most frequent /eyC/ words by date of birth. Source: Labov 2017.

only once in the corpus of language change? Zipf's Law (1949) tells us that a great majority of words will be used sparingly. There are 1,650 /eyC/ words in the PNC data: only 56 appear more than 100 times, and there are 662 singletons, for which it is impossible to carry out a trend analysis.

Upon reflection, the sparsity problem is not limited to historical corpus where the dataset is necessarily finite. In any study of language variation where a word may take on multiple forms, there is no guarantee that a particular form will be attested just because it can be attested. To see this, let us consider a stable linguistic variable, the deletion of t/d in English word-final consonant clusters.¹ A word such as *hand* may surface with or without the final /t/. It appears in the Buckeye Corpus 41 times: the rate of deletion is 39% and the rate of retention is 61%. However, 447 out of the 1,083 Buckeye words eligible for t/d-deletion appear only once. In a smaller dataset, one developed from the Penn LING560 fieldwork in the Philadelphia speech community, 134 of the 288 words are singletons. Figure 2 gives the distribution of words by deletion rate from the two corpora.

The majority of word (types) in these corpora actually show no variation; they either categorically delete or categorically retain t/d. More precisely, only 76 out of the 288 words from the LING560 corpus, and only 416 out of the 1083 words from the Buckeye corpus, show variation; that is, there are plenty of non-singletons that do not show variation either.

The plots in Figure 2 would be characteristic of all cases of (probabilistic) linguistic variation. As such, the linguist will never be able to give a complete description of the variation, perhaps never satisfying a reviewer who — unreasonably in our opinion — insists on proof of neogrammarian regularity for singletons. At the same time, there is no doubt that variable processes *are* acquired by children, and often remarkably early and accurately when measured against the speech community at large (Labov 1989, Roberts and Labov 1995, Roberts 1997, Smith et al. 2009, Miller and Schmitt 2012, Nardy et al. 2014). The data sparsity problem raises a sharp challenge for the language learner: How would one learn that the *variable* process such as t/d-deletion is in fact *categorical*, in the sense that it applies to all eligible words? This first step of

¹We thank Josef Fruehwald for providing us with the t/d-deletion data from the Buckeye Corpus.

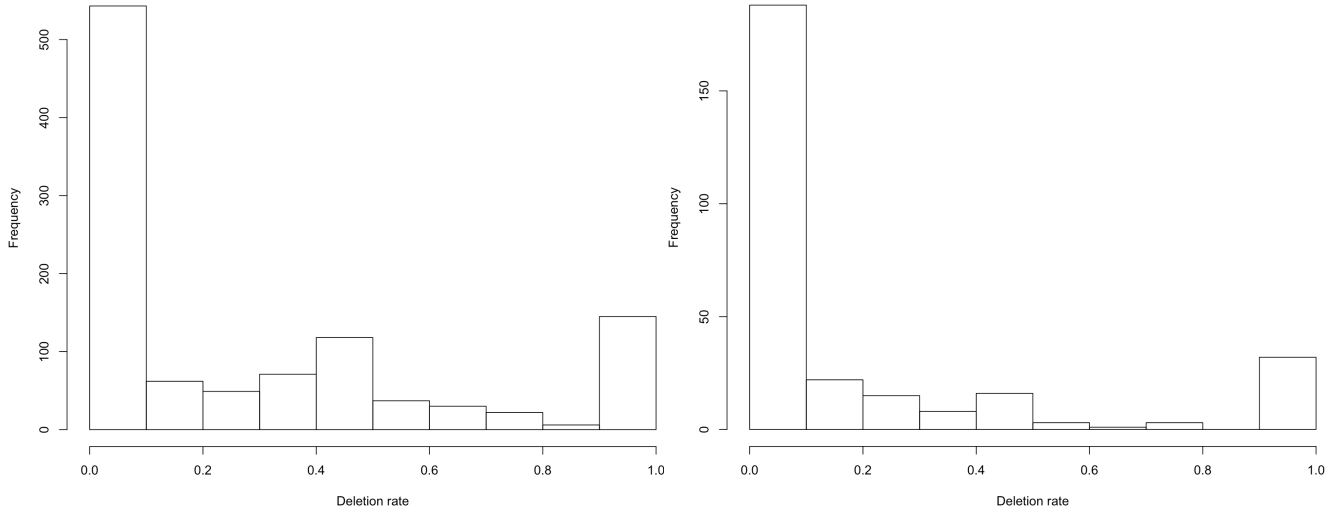


Figure 2: Histogram of t/d-deletion. Left: Buckeye Corpus (courtesy of Josef Freuhwald); Right: LING560 Corpus.

categorical acquisition must logically precede the gradual approximation of probabilistic usage frequencies in the speech community. To do so, an important step of generalization must take place, as we discuss presently.

3 A Learnability Proposal

Let us consider how a child learner may acquire variables in a hypothetical example. A variable rule is defined over a set of N items and yields two forms, A and B , with probability of p and q respectively. If the rule were to be used infinitely many times, then each of the N items will appear in form A as well as form B , with probability p and q . But of course no one talks forever. Given the statistical distribution of linguistic items (Zipf 1949), the data generated by the variable rule must consist of:

- (1) a. A very small fraction of the N items that appear in both form A and B , more or less reflecting the underlying probabilistic distribution of p and q ;
- b. A significant subset of the items which appear in both form A and B but have usage probabilities quite different from p and q ;
- c. A large subset of items that appear exclusively in form A or B , including many that are used only once and therefore cannot show variation.

If the statistical distribution described in (1) is representative of naturalistic data, then the acquisition of probabilistic rules requires a very significant step of generalization. The learner must conclude that the behavior shown for a tiny fraction of the lexical items is to be extended to the entire set, including even those unattested in the learning input.

Similar situation arises in the acquisition of non-probabilistic aspects of language, which are also “under-represented” in the data. In earlier work, we have investigated the status of syntactic rules in early child English (Yang 2013). A fully productive rule “NP→D N” suggests that the determiner ($D = the$ and a/n) can be interchangeably used with singular nouns ($N = car, dog, child, etc.$). In numerous corpus analyses, both children and adults produce fairly low values of combinatorial diversity: typically only 20-40% of nouns are

paired with both determiners (Pine and Lieven 1997, Valian et al. 2009). Yet a rigorous statistical test shows that even very young children reach the level of diversity which, while low, is expected under a categorical rule that independently combines determiners and nouns (Yang 2013; see Silvey and Christodoulopoulos 2016, Meylan et al. 2017, Zaccarella 2018 for independent replications).

Consider a concrete case. Adam, an American-English-learning child studied by Roger Brown (1973), produced 3,729 determiner-noun combinations in his speech with 780 distinct nouns. Of these only 32.2% appeared with both determiners, which is similar to the expected value of 33.7% under the abstract “NP → D N” rule. Adam’s mother, who was recorded in the same corpus, produced a diversity measure of 30.3% out of 914 nouns. Even among the 469 nouns used at least twice, which provided opportunities to be used with both determiners, only over half (260) did so. A giant leap of faith is needed: Adam must — and apparently did at a very young age — generalize from a small subset of nouns with attested interchangeable determiners to all nouns.

Note that the logic of the NP rule acquisition is exactly the same as the acquisition of variable rules: the learner must conclude from a small set of items that show variation that all items can be variably used. Here we propose a formal solution of how to acquire variation, broadly construed to encompass both probabilistic and categorical aspects of language. We draw upon the Tolerance Principle (TP; Yang 2005, 2016), a learning theory that identifies productive processes in language acquisition.

- (2) Let a rule R be defined over a set of N items. R is productive if and only if e , the number of items not supporting R , does not exceed θ_N :

$$e \leq \theta_N = \frac{N}{\ln N}$$

The TP builds on the intuition, shared by many (e.g., Aronoff 1976, Plunkett and Marchman 1993, Bybee 1995), that rules must “earn” productivity by the virtue of being applicable to a sufficiently large number of candidates it is eligible for. If there are 10 examples and all but one (9/10) support a rule, generalization ought to ensue. But no one in their right mind would extend a rule on the basis of 2/10: the learner should just memorize the two supporting examples. Productivity is a calibration of regularities and exceptions — crucially with respect to word *types* rather than tokens. If e exceeds θ_N , then the learner will “lexicalize” only these and not generalize beyond them: that is, R is unproductive.

A rule and its numerical values are clearly subject to individual variation: if speakers differ from each other in anything, it is their vocabulary. It is possible for some learners to detect the productivity of a rule earlier than others, and it is also possible for the productivity of a rule to change during the course of language acquisition: they are all governed by the fluctuation in an individual’s vocabulary composition, namely N and e ; see (Yang 2016, Chapter 4) for case studies. Should these differences persist into adulthood, they will introduce heterogeneity in the linguistic data for future learners, thereby creating potentials for language change. At the same time, the calibration of productivity under the TP deals with the type frequency of words, and more specifically, the proportion of e relative to N . It is immaterial exactly what these words are, or how frequently they appear — assuming, of course, they are frequent enough to be learned by young children. Therefore, while different speakers will necessarily know different words, for obvious and non-obvious reasons, their grammar may still be the same if the relative proportions of N and e fall on the same side of productivity. This may account for the general finding of uniformity in the outcome of language acquisition throughout a speech community (Labov 1972).

A useful feature of the TP lies in its ability to detect linguistic generalizations, a task that both child learners and linguistic theorists have to face. One possible outcome is that as long as a rule R is assessed to be productive, i.e., the number of exceptions falling below the threshold, there may be no “deep” motivation for the nature of these exceptions. For instance, while all theories of English morphology must recognize

the productivity of the regular past tense suffix “-ed”, it would be folly to require the theory, at least in the synchronic sense, to predict the exact set of irregular verbs as they are residues of historical contingencies. Another possible outcome is that the rule is unproductive, i.e., the number of exceptions exceeding the threshold: the learner will resort to memorizing only those items that do follow the rule, on the basis of positive attestation, and there again may be no rhyme or reason why *these* items follow the rule but not others. An example here would be the English ditransitive verbs of Latin origin: only a small fraction (e.g., *assign*, *offer*, and *promise*) can participate in the double object construction while the vast majority (e.g., *donate*) do not (Yang 2016). The learner will simply rote-learn those that are positively attested: because the productivity threshold is not met, they will generalize beyond the input.

The TP provides a concrete model for the study of language change: when the quantitative data is available, the TP gives precise predictions when a rule gains or loses productivity, and thereby gaining or losing lexical items under its reign; see Ringe and Yang (2018) and Sneller et al. (2018) for two recent applications. Here we provide a brief account of the variation and change in the auxiliary selection system of Montreal French as discussed by Sankoff in a series of publications (Sankoff and Thibault 1977, Sankoff 2019).

Romance and Germanic languages show considerable variation in the choice of the auxiliary (roughly *be* vs. *have*) with intransitive verbs. While some tendencies can be identified with respect of the semantic properties of the verbs (Levin and Hovav 1995, Sorace 2000), they are by no means categorical (Rosen 1984); indeed, there has been considerable diachronic changes in auxiliary selection in the history of these languages (e.g., Vincent 1982, Aranovich 2003). In the case of modern French, those that select *être* (‘be’) are verbs of motion, state, and change of state but they form an idiosyncratic list of fewer than 20 items, which students frequently rely on mnemonic devices (“Dr. Mrs. Vandertramp”) to memorize. Because the total number of verbs in these semantic classes is much larger, the default/productive auxiliary would clearly be *avoir* (‘have’). The fact that the *être* verbs are the exceptions can be seen in their rates of variation in usage and frequencies (Sankoff 2019, Table 4, reproduced in part below in Table 1).²

Verb	Gloss	Percent <i>avoir</i>	Total
<i>monter</i>	go up	76.4%	55
<i>passer</i>	pass by	75.3%	93
<i>déménager</i>	move [to a new home]	71.8%	163
<i>tomber</i>	fall	66.9%	172
<i>rester</i>	stay	66.7%	531
<i>descendre</i>	come down	60.6%	33
<i>entrer/rentre</i>	(re)enter	53.7%	203
<i>sortir</i>	go out	52.7%	205
<i>partir</i>	leave	32.4%	398
<i>retourner</i>	return	43.9%	98
<i>partir</i>	leave	32.4%	398
<i>arriver</i>	arrive	8.6%	997
<i>venir</i>	come	7.3 %	618
<i>revenir</i>	come back	1.5%	206
<i>aller</i>	go	0.1%	697

Table 1: The variable use of *avoir* ‘have’ in Montreal French. Adapted from Sankoff (2019).

²The three change-of-state verbs (*devenir* ‘become’, *naître* ‘be born’, and *mourir* ‘die’), which do not show variation, are excluded. The morphologically related verbs *entrer* (‘enter’) and *rentrer* (‘reenter’) are merged.

The rate of *avoir* use is significantly negatively correlated with lexical frequency ($r = -0.68$, $p = 0.005$). This is a hallmark of lexicalization: because the *être* verbs need to be rote-learned, less frequent verbs are more likely to succumb to the pull of the productive form, much like how low frequent irregular verbs are more likely to be regularized during language acquisition and change (Marcus et al. 1992, Yang 2002). The only route to resistance is repeated exposure: indeed, Sankoff (2019) shows an increase in the use of *être* over the past 100 years or so, which is accounted for the speakers’ rising level of education and literacy.

Turning to another useful feature of the TP with respect to the acquisition of variation, Table 2 provides some sample values of N and the associate threshold values θ_N , the maximum number of exceptions that a productive rule can tolerate.

N	θ_N	%
10	4	40.0
20	7	35.0
50	13	26.0
100	22	22.0
200	38	19.0
500	80	16.0

Table 2: The maximum number of exceptions for a productive rule over N items.

The TP asserts that a smaller vocabulary (i.e., smaller values of N) can tolerate a higher percentage of exceptions: all else being equal, productive rules are easier to detect for learners who have access to less input data. In other words, if the learner ignores words that are not sufficiently frequent—as in the beginning stages of language acquisition (Fenson et al. 1994, Goodman et al. 2008)—, then the odds of acquiring a general rule improve considerably. This property has important consequences for the learnability of language.

Returning to the case of determiner-noun combinations, we see that in the speech transcript of Adam’s mother, 43 of the top 50, 87 of the top 100, and 124 of the top 150 most frequent nouns are used with both determiners. These are very high batting averages for the “NP→D N” rule, all clearing the productivity thresholds under the TP, which are at 37, 78, and 120, respectively. Because English-learning children acquire the fully productivity of the NP rule very early (prior to age 2), their vocabulary size is very small and consists of probably no more than 500 words at the maximum (Hart and Risley 1995, Rowe and Goldin-Meadow 2009, Rowe et al. 2012): precisely the region of N for which the evidence for the interchangeability of the determiners is robustly productive.

It is also clear that the “small-is-better” character of the TP will facilitate the acquisition of probabilistic linguistic variation. More frequent items are more resistant to sampling effects and are thus more likely to reflect the full range of variation. With respect to the t/d-deletion process, we see that in the LING560 corpus, 43 of the top 50 most frequent words show both deletion and retention of t/d. In the larger Buckeye Corpus, all of the top 50, 99 of the top 100, and 192 of the top 200 words are variable. Such statistical patterns are likely to hold for language acquisition data. For example, Roberts (1997) shows that Philadelphia children have acquired most of the phonological and morphological constraints on t/d-deletion by the age of three, including crucial aspects of the dialect features (c.f., Guy and Boyd 1990). A larger study comparing child and caretaker speech data (Smith et al. 2009) replicates the findings on the structural factors with the additional demonstration that children’s language is inherently probabilistic without lexical effects. Again, children in this age range have no more than about 1,000 words in their vocabulary altogether (Hart and Risley 1995). To get a rough approximation of the t/d-deletion process in the input, we extracted the top 1,000 most frequent words in a six-million-word corpus of child-directed corpus (roughly a year worth of input

data for high SES children and two years for low SES children). Of these, 90 words meet the structural conditioning for t/d-deletion. Even in the much smaller, 400,000-word, Buckeye corpus, 75 of these are used variably — which already justifies the generalization that *all* eligible words are subject to variable t/d-deletion as only 70 would be sufficient ($\theta_{90} = 20$). Very young children, then, can easily detect t/d-deletion as a general rule of their language. Here it is significant that children do *not* match the deletion rate of their caretakers in Smith et al.’s study. Presumably, the matching of deletion probabilities will take place gradually over the later stages of language development: in other words, children learn the fact of variation as a categorical rule first from a small but highly frequent sample for which variability is abundantly attested, as sanctioned by the TP. Once this generalization is made, that words of specific structural properties *can* delete the word-final t/d, the learner will extend this generalization to all words in the class, including low frequency words that do not show variation themselves as well as words that they have not learned. The pattern of probability matching can be acquired by tracking the frequencies of the variants — deletion vs. retention — *irrespective* of the lexical item.³

4 Transmission and Diffusion Redux

The differences between child and adult language acquisition have been well documented with special focus on the critical period hypothesis (Johnson and Newport 1989, Scovel 2000, DeKeyser 2000), which we have sought to distinguish in language variation and change (Labov 2007, Sankoff 2018). A key empirical study is the diffusion of the New York City and mid-Atlantic short-a system to various dialectal regions across the US. The complex structural conditioning factors, some of which are abstract while laden with lexical exceptions, were quickly lost as the result of imperfect adult language learning. By contrast, the major components of the (almost) equally complex Philadelphia short-a system be acquired accurately by children as young as three (Roberts and Labov 1995, Labov 1994): they consistently tense short a before nasals and syllable boundaries but never do so for short a in initial positions (e.g., *animal*), closely matching the adult community form.

We suggest that the learnability model of linguistic variation based on the TP provides a unified account for transmission and diffusion. The basic proposal is that the TP is operative for *both* child and adult language acquisition, but the critical difference between these learners lies in the state of their lexicon: in particular, the size of the vocabulary (N in the TP calculation) relevant for the acquisition of linguistic rules (Yang 2018). There is now converging evidence that the lexicons in multilingual speakers are continuously activated (see Jiang 2000, Dijkstra 2005, Tokowicz 2014, and Kroll and Ma 2018 for reviews). If so, adult learners of a second language or dialect must accumulate evidence for grammatical rules on top of their native vocabulary, which is already sizable: as shown in Table 2, a larger N makes it much more difficult to detect productive rules given the data sparsity problem. Children, by contrast, have the advantage of starting afresh with a small lexicon, for which it is easier to identify rules as discussed in Section 3.

The validity of our proposal can only be established upon further study; here we can only offer some suggestive preliminary evidence. Most fundamentally, how do we show that the TP is used by adults at all, when there is *prime facie* evidence that suggests otherwise? For example, in a series of studies (summarized in Schuler 2017), subjects learn artificial languages where rules have various levels of exceptions. Children follow the TP nearly categorically but adults generally match the token frequencies of the available forms in the stimuli as shown in previous work with a somewhat different task (Hudson Kam and Newport 2005, 2009). A straightforward account for the differences between child and adult language learning would to postulate that the TP as a learning mechanism is available to only child learners.

³Such a learning model is not incompatible with lexically specific effects such as frequency: certain items may exceptionally deviate from the target probabilistic distribution (e.g., Walker 2012).

However, at the present time we cannot rule out the child-vs.-adult difference in experimental learning situations as a task effect, rather than a fundamental change in their mechanisms of language learning. On the one hand, that adults probability match while children prefer categorical rules can be found in other domains of learning and decision making (e.g., Weir 1964, Derks and Paclisanu 1967). On the other, we know since (Berko 1958) that adults approach linguistic experiments differently from children; see Armstrong et al. (1983) and Schütze (2005, 2011) for related discussion across domains and tasks. In the original set of Wug tests, the subjects were presented with novel verbs such as *spow* and *bing*, which are similar to existing irregular verbs (*blow-blew*, *sing-sang*). Only one of the 86 children tested produced bang (no said *spew*, and only one *spowed* was produced). Adults, however, are far more willing to produce irregularized forms, which is likely to a task effect, since there have been virtually no such cases in the natural history of English verbs (Anderwald 2013, Yang 2016).

We contend that there are at least two reasons to believe that the TP holds for learners of all ages. Theoretically, the derivation of the TP requires only certain assumptions about how rules and their lexical exceptions are represented (Yang 2016, Chapter 3), which are not known to degrade across language development. In fact, the psycholinguistic motivation for the TP is established in both child and adult populations (e.g. Clahsen et al. 2004, Fleischhauer and Clahsen 2012). Empirically, there is evidence that the window of first language acquisition must remain open for sufficiently long such that older children and indeed L2 adults can learn rules extremely well.

With respect to L1 acquisition, we should not overlook the late acquisition of many specific rules and constructions in spite of children’s general mastery of language. For instance, a significant portion of English derivational morphology is acquired quite late (Tyler and Nagy 1989, Jarmulowicz 2002), presumably because it involves advanced vocabulary that comes with literacy and education. The complex distribution of the dative constructions are also acquired late (Gropen et al. 1989), as children persist in producing errors such as “I said them no” and “I delivered you lots of pizza” well after the core grammar is firmly in place. And the classic work of Chomsky (1969) shows that the constructions closely tied to lexical development such as raising, control, and “tough” movement emerge very late in first language acquisition. With respect to L2 acquisition, evidently adult learners *can* form productive rules in a manner similar to L1 learners. Yang and Montrul (2017) provide an extensive review of L2 acquisition of the English dative constructions by adults whose L1 do not have these constructions. These constructions offer a natural case study of L2 learning “in the wild” because (a) they are language specific and cannot be attributed to universal principles of grammar (e.g., White and Genesee 1996), and (b) their grammatical properties are obscure and are unlikely to be under the influence of formal instruction — ESL teachers would never offer lessons on the distribution of *donate*. Nevertheless, L2 learners go through the stages of over-generalization and retreat like L1 learners, and they gradually refine the phonological and semantic restrictions on the constructions over time, with advanced learners showing native-like grammaticality judgment (e.g., Jäschke and Plag 2016).

If some grammatical rules can be learned near perfectly, one wonders why most aspects of second language are not native-like. As reviewed earlier, very young English-learning children know that the combination of determiners and nouns is a productive rule of their language. Using the statistical benchmark developed in Yang (2013) on a publicly available second language acquisition corpus (Perdue 1993), it can be shown that these adult learners evidently do not have the productive use of the determiner rule (Yang 2018), supporting the conclusion from previous behavioral analysis of L2 English (Liu and Gleason 2002, Ionin et al. 2008, Snape 2008).

The likely reason for the discrepancy between adults’ eventually successful acquisition of a very complex construction (e.g., the datives) and their evident failure to grasp a very simple rule (e.g., “NP→D N”) lies in the vocabulary size involved in their acquisition. The verbs that participate in the dative constructions have specific semantic properties and essentially form a closed set: it is unlikely for native English learners to encounter more than 100 such verbs (Gropen et al. 1989, Yang 2016). This affords an L2 learner the

opportunity to relive the process of L1 acquisition by building a set of semantically cohesive verbs, from ground up, that participate in a syntactic construction. For the “NP→D N” rule, however, an L2 learner already has a huge L1 inventory of nouns to begin with, which places them at a disadvantage when it is necessary to detect linguistic regularities in a sparsely distributed input sample. The same would be true for the acquisition of word order, inflectional morphology, and other core components of language, precisely where L2 learners struggle, as they have to contend with the influence from their L1 lexicon. If so, adults would struggle to acquire the complex short-a NYC system (Labov 2007): they already have a full lexicon albeit with a different phonology and their input is necessarily reduced, and more sparse as a result, in the situation of dialect contact (Sneller et al. 2018). Note that this leaves open the possibility that adults may be able to acquire item-specific features of a second dialect, as noted in Payne’s 1976’s study of out-of-state families acquiring the Philadelphia system in King of Prussia.

The last piece of the argument concerns when rule learning a la the TP stops. Since the TP requires a great majority of words to follow a rule to ensure productivity, waiting too long before coming to a decision (i.e., with a large N) would render every rule unproductive because of the data sparsity as discussed in the earlier sections. The answer comes in two ways, both suggesting that the learner will stop looking and “freeze” the rules in place at a value of N no more than a few hundred. Recall a three-year-old’s vocabulary size is no more than just around 1,000 (Hart and Risley 1995): productivity decisions can, and must, be made when N is quite modest. It is important to stress that the learning limit of N is not a function of age. As noted earlier, the full details of some language specific constructions are learned quite late but the cardinality of the relevant words (e.g., ditransitive verbs) is always very limited. It seems that as long as N remains at a modest value, the learner will keep their options open. Why this may be so is currently mysterious. We know nothing about how something like the TP is implemented computationally never mind neurologically. It is extremely unlikely if learners could report back how many rule-following and rule-defying items they have learned during a brief experiment (Schuler 2017) even though their behavior follows the numerical predictions of the TP very closely. And it is virtually certain that the values of N , e , and θ_N can not be precisely represented, especially if they are large.

To summarize, we have suggested that a learning principle such the TP may be necessary to overcome the sparse data problem in the acquisition of linguistic variation. Moreover, its elevated effectiveness when applied to smaller lexicons may shed light on the patent differences between child and adult language acquisition and provide a psychological underpinning for transmission and diffusion in language change.

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